

Description of Requirement (DoR)

Electro-Magnetic Gradiometer Sensing System Hardware (EMG-SSH) Device

Specifications:

The U.S. Army Engineer Research and Development Center (ERDC) Geotechnical and Structures Laboratory (GSL) seeks purchase of one (1) Electro-Magnetic Gradiometer Sensing System Hardware (EMG-SSH) device that is sensitive between 50kHz-750kHz and can make measurements in the picotesla range to enable detection and mapping of long linear subsurface conductors.

System Description:

The ERDC GSL has the need to procure one (1) rugged Electro-Magnetic Gradiometer Sensing System Hardware (EMG-SSH) Device with the following specifications:

- Rugged yet lightweight man-carry boom that can be disassembled and fit into an included 32in x 21in x 13in pelican case, or other similar rugged case, with packing foam designed to securely hold the components for storage and shipping.
- Man-carry boom of about 1.75m (5.75 feet) long and including fittings for the rigid mounting of receiving antennas at each end (fore and aft)
- Man-carry boom designed to hang from straps attached to a backpack, weight balanced front to rear and left to right when all components are attached
- Man-carry boom designed to hang in-line with users walking direction and centered on the user's body with angled portions that continue the boom around the side of the end-user's body
- Man-carry boom ergonomically designed, and weight balanced to have little to no pitch, yaw, or roll when end-user is walking with the backpack supported instrumentation
- Differential EMG receiver electronics, sealed to protect from moisture and dust, mounted on the man-carry boom for recording magnetic field measurements across a large range of field intensities (many orders of magnitude) from large primary fields from transmitters to much smaller secondary fields in the pico-Tesla range from reradiating targets
- Differential Receiver electronics with a minimum of two (2) channels programmable within the 50kHz-750kHz range to 10Hz accuracy mounted to the boom near the center in front of the user
- Differential EMG receiver electronics obtains power from external Bren-Tronics model BB-2590/U, 9.9 AH Rechargeable Lithium-Ion Battery
- Two (2) Bren-Tronics model BB-2590/U, 9.9 AH Rechargeable Lithium-Ion Batteries and two (2) matched chargers
- Matched electrically shielded Magnetic Dipole Antenna pairs, each antenna comprised of a metal grillwork wrapped around a central ferrite core loop antenna; wherein, the metal grillwork is longitudinally split to prevent shorting out signals detectable by the central ferrite core loop antenna, which are

tuned in the 50kHz to 750kHz range and are no longer than 18 inches – to obtain required tuning across the frequency range of interest it is expected that 2 or 3 matched pairs of Dipole Antennas may be necessary.

- Magnetic Dipole Antenna pairs should be capable of detecting secondary fields generated in long conductors (>100m) by non- licensed intentionally radiating transmitters operated 10% down from the Federal Communications Commission (FCC) maximum emission limits as defined in CFR 2009 Title 47 Vol. 1 Part 15 section 209.
- Dipole Antenna pairs designed so that they can be attached at each end of the boom, in either horizontal or vertical orientation based on situational need, that are sensitive to fields in the picotesla range, as described in the previous bullet, each electrically connected to a respective and separate input on the Differential EMG receiver
- Dipole Antenna pairs are to be connected to the differential EMG receiver electronics by cables with relevant Mil-Spec connectors (examples of appropriate connectors can be seen on <https://www.milnec.com/mil-spec-connectors/>) providing for protection from moisture and dust
- Independent and apart from the Differential EMG receiver, a small tablet computer device mounted on the man-carried boom connected to the Differential EMG receiver by cable with the ability to download stored data via removeable storage device(s) (USB, SD Card, or other similar)
- Data Acquisition and Display Software (SW) installed on the small tablet computer that enables end-user to choose and perform frequency setup and validation, control initiation and stoppage of data collection, both store and visually display to a user the relative locations and magnetic field measurements collected by the differential EMG receiver in real-time as data are being collected and saved to the device
- Data Acquisition SW will save EM data from each receiver independently, front and rear, at a fixed sampling rate of no less than 10Hz
- Data Acquisition SW will provide indication of current filename, system status alert (green – all is good, yellow – something requires review, red – something is not operating properly) – status should include monitoring of battery, internal electronics temperature, GPS status
- Differential EMG receiver electronics has a port for, accepts data from, and merges GPS navigation data, which are used to locate and log the changing positions of the EMG receiver, with the recorded EMG data – note merging of EM data and GPS data can occur within the Differential EMG Receiver electronics or in the Data Acquisition and Display Software on the tablet computer
- Automatic file naming and saving using MMDD_###, where MM is two-digit month, DD is two-digit day on the month, and ### is three-digit sequential file number for that day based on start/stop control in Data Acquisition and Display Software
- Saved and output data file format that meets the following criteria; csv file format with a header in the data file that indicates: Number of header lines (or some indicator of start/end of header), System label (e.g., ACME EMG v1.1), RX mode (horizontal or vertical), Any other relevant details, gains, filters, etc., Date/time of collect (start or end), Channel 1 = frequency F1, Channel 2 =

frequency F2, ..., Channel N = frequency FN. Following the header file should be receiver data that is time synchronized for all channels acquired (double-precision) and merged with incoming 1HZ GPS data; Output data frequency should follow the below table and be no less than 10 Hz; Channel 1 is the first frequency, Channel N is the Nth frequency, RX1 is the front RX and RX2 is the rear RX;

	Channel 1				...	Channel N					
Time	RX1 real (inphase)	RX1 imag (quadrature)	RX2 real (inphase)	RX2 imag (quadrature)	...	RX1 real (inphase)	RX1 imag (quadrature)	RX2 real (inphase)	RX2 imag (quadrature)	latitude	longitude

Or,

	Channel 1		...	Channel N			
Time	RX1 complex	RX2 complex	...	RX1 complex	RX2 complex	latitude	longitude

- A pair of staging tripods or similar with mounting clamps for supporting the man-carry boom when not in use, facilitate the assembly, disassembly, and configuration changes (preparation for field use), and assist the user in positioning themselves under backpack shoulder straps
- Backpack with support straps that are attachable to the boom allowing hands-free carrying and operation
- Mobile (non-RTK, must work solely from satellites without need for a base station) GPS navigation receiver (Trimble SPS 852 or similar) with the ability to obtain sub 20cm accuracy to include rugged case for storage and shipping
- GPS navigation receiver to fit within backpack, GPS antenna to be attached to but external to the backpack on a mounting pole
- Bren-Tronics Battery to attach to backpack or fit within backpack with power cord that attaches to the Differential EMG receiver electronics
- Electro-Magnetic Gradiometric Sensing System Hardware (EMG-SSH) as described in this document, to include all components required for data collection (Man-carry boom with small tablet computer, backpack, Bren-Tronics battery, GPS, Differential EMG receiver, Dipole Antenna pairs, associated cables, etc.) should not exceed 30 lbs.
- Operator's and User Guides describing system assembly and operation
- A rugged or semi-rugged laptop (Panasonic FZ-55 or similar), corei7, 16GB RAM, 250GB HD, data input capability consistent with output method from small tablet on EMG (USB, SD Card, or other similar),
- One (1) additional 32in x 21in x 13in Pelican case or similar rugged case with packing foam designed to hold all remaining components for storage and shipping.

Delivery/Acceptance:

Complete Device must be delivered and determined acceptable by government personnel no later than 6 months after award or agreed upon alternative delivery date.

In order to be determined acceptable the device shall be tested by government personnel to ensure the system meets 100% of the specifications as stated within the DOR and is 100% functional when tested as stated below.

****The contractor shall not receive payment until the item has been delivered, tested, and determined acceptable by government personnel.**

The EMG-SSH will be tested for the ability to detect secondary fields generated in long conductors (>100m) by non- licensed intentionally radiating transmitters (frequency range in the DoR). These will be operated 10% down from the FCC maximum emission limits as defined in CFR 2009 Title 47 Vol. 1 Part 15 section 209. This testing will be completed at distances of 10m and greater. Failure to meet the 10m threshold will be considered not meeting the technical requirements and will be rejected until this requirement is met.

Deliverables:

The contractor shall provide the following deliverables:

- One (1) full system as defined in the DOR; including peripheral devices (GPS, instrument operation tablet and stand-alone computer)
- System Design Drawings (both physical and electronic-pdf)
- Instrument assembly/disassembly manual (both physical and electronic-pdf)
- Instrument operation manual (both physical and electronic-pdf)
- Software operation manual (both physical and electronic-pdf)
- In-person training on instrument assembly, instrument operation at Vicksburg, MS

Delivery Location: Engineer Research and Development Center
3909 Halls Ferry Rd.
Geotechnical and Structures Laboratory
Bldg. 3296
Vicksburg, MS 39180

***Please note: Shipping terms shall be FOB Destination. All shipping costs shall be included within the vendor's quote.**